

Math 2270 Quiz 1 Key

6pts

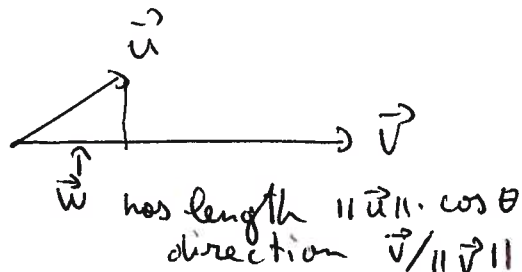
① Find the angle between $\vec{u} = (1, 2, 0)$ & $\vec{v} = (3, -1, 1)$ and the projection of $\vec{u}$ onto $\vec{v}$.

$$\vec{u} \cdot \vec{v} = \|\vec{u}\| \|\vec{v}\| \cos \theta$$

$$3 - 2 + 0 = \sqrt{5} \sqrt{11} \cos \theta$$

$$\cos \theta = \frac{1}{\sqrt{55}}$$

$$\theta = \cos^{-1} \frac{1}{\sqrt{55}}$$



$$\vec{w} = \text{proj}_{\vec{v}} \vec{u} = \frac{\vec{u} \cdot \vec{v}}{\|\vec{v}\|^2} \vec{v} = \frac{1}{11} \cdot (3, -1, 1) = \left(\frac{3}{11}, -\frac{1}{11}, \frac{1}{11}\right)$$

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② Solve $\begin{cases} 2x + 4y + 4z = 6 \\ -x - 2y - z = -1 \\ 2x + 4y + 7z = 12 \end{cases}$

What shape do the solutions represent?

$$\begin{array}{ccc|c} 2 & 4 & 4 & 6 \\ -1 & -2 & -1 & -1 \\ 2 & 4 & 7 & 12 \end{array} \rightarrow \begin{array}{ccc|c} 1 & 2 & 2 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 3 & 6 \end{array} \rightarrow \begin{array}{ccc|c} 1 & 2 & 0 & -1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{array}$$

$$z = 2 \quad x = -1 - 2y \quad \begin{pmatrix} -1 - 2y \\ y \\ 2 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix} + y \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}$$

A line of solutions.

Notes: Median = 8
High = 11